

Table 1: Sequential MAS results for Qwen3-8B. Acc. is accuracy (%). Single, TextMAS, and Linear times are converted to the Kernel machine’s throughput. Best results in each row are bold and second-best results are underlined. Kernel and Kernel-ES are highlighted. Improve compares the better of Kernel and Kernel-ES with TextMAS.

Tasks	Metrics	Baselines		LatentMAS variants				Improve
		Single	TextMAS	Linear	Kernel	Kernel-ES	Soft	
Sequential MAS Setting								
AIME24	Acc.	50.0	53.3	56.7	66.25	<u>64.17</u>	63.75	↑ 13.0
	Token	12891	38596	8953	8548	<u>8770</u>	22342	↓ 77.9%
	Time	340.2	1018.6	236.3	225.6	<u>225.8</u>	549.0	× 4.5
AIME25	Acc.	46.7	53.3	53.3	49.58	<u>50.42</u>	49.17	↓ 2.9
	Token	14692	45088	8699	10158	<u>9830</u>	24557	↓ 78.2%
	Time	377.8	1159.4	223.7	261.2	<u>255.5</u>	597.0	× 4.5
GPQA	Acc.	39.9	43.4	45.5	<u>48.74</u>	48.61	49.24	↑ 5.3
	Token	6435	17986	4571	3588	<u>3846</u>	13076	↓ 80.1%
	Time	104.6	292.2	74.3	58.3	<u>59.4</u>	257.6	× 5.0
HumanEval+	Acc.	74.4	80.5	80.5	83.54	<u>83.38</u>	82.62	↑ 3.0
	Token	2507	4593	1866	1637	<u>1696</u>	15055	↓ 64.4%
	Time	31.5	57.7	23.4	18.8	<u>21.5</u>	112.4	× 3.1
MBPP+	Acc.	64.8	69.5	74.6	75.26	<u>75.20</u>	73.15	↑ 5.8
	Token	2053	3695	1164	<u>1462</u>	1528	14388	↓ 60.4%
	Time	28.9	52.1	16.4	<u>17.9</u>	19.6	101.6	× 2.9
MedQA	Acc.	53.0	<u>75.0</u>	75.3	73.67	72.00	66.67	↓ 1.3
	Token	2098	4260	1555	1139	<u>1154</u>	15972	↓ 73.3%
	Time	18.3	37.1	<u>13.5</u>	12.5	13.6	111.1	× 3.0

Note. MedQA may be removed. Repetition rule: datasets with 1–100 problems are repeated 8 times; those with 101–1000 problems, 4 times; and those with more than 1000 problems, once.